

## Gradient function graphs

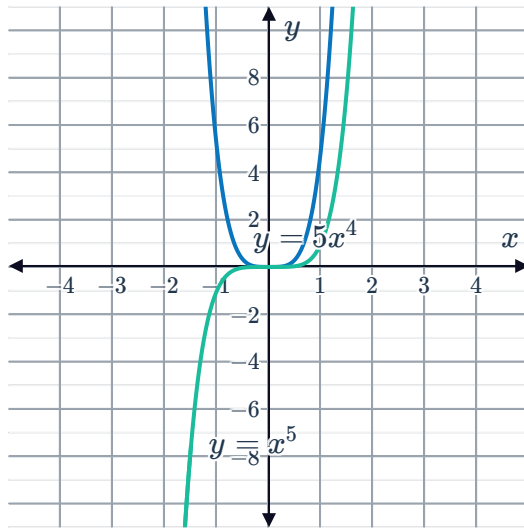


- 1
  - a 1
  - b 1
  - c A linear function has a constant gradient.
  
- 2
  - a The gradient function is positive when  $y = x^2$  is increasing, and negative when  $y = x^2$  is decreasing.
  - b Positive
  - c The gradient of the tangent line increases at a constant rate.
  - d Negative
  - e The gradient of the tangent line increases at a constant rate.
  - f Linear
  
- 3
  - a Positive
  - b The gradient of the tangent line increases at an increasing rate.
  - c Positive
  - d The gradient of the tangent line decreases at an decreasing rate.
  - e Quadratic
  
- 4
  - a Positive
  - b The gradient of the tangent line increases at an increasing rate.
  - c Negative
  - d The gradient of the tangent line increases at a decreasing rate.
  - e Cubic

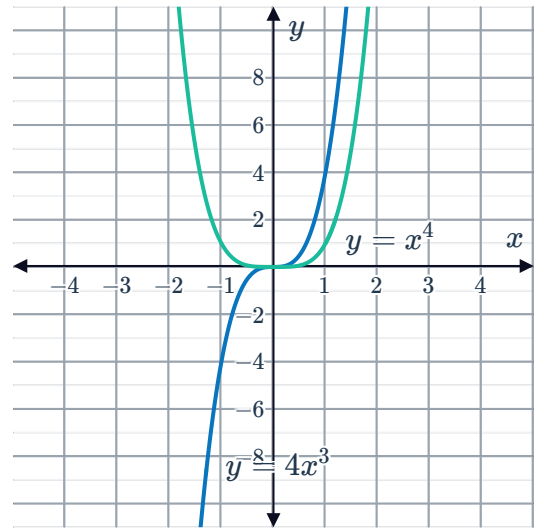
5



a



b



c Odd

d Even

## Algebraic differentiation

6

a  $2x$

b  $8x^7$

c  $nx^{n-1}$

7

a

i Degree 1

ii Degree 2

iii Degree 3

b  $n - 1$

8

a  $\frac{dy}{dx} = 7x^6$

b  $\frac{dy}{dx} = 9x^8$

c  $\frac{dy}{dx} = -7x^{-8}$

d  $\frac{dy}{dx} = -x^{-2}$

e  $\frac{dy}{dx} = \frac{6}{5}x^{\frac{1}{5}}$

f  $\frac{dy}{dx} = \frac{2}{3}x^{-\frac{1}{3}}$

g  $\frac{dy}{dx} = 0$

h  $\frac{dy}{dx} = 0.6x^{-0.4}$

9

a  $y = x^{\frac{1}{2}}$

b  $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

10

a  $y = x^{-2}$

b  $\frac{dy}{dx} = -2x^{-3}$

11

a  $y = x^{-\frac{1}{4}}$

b  $\frac{dy}{dx} = -\frac{1}{4}x^{-\frac{5}{4}}$



12  $\frac{dy}{dt} = \frac{3}{2}t^{\frac{1}{2}}$

13  $\frac{dy}{dx} = -\frac{1}{5x^{\frac{6}{5}}}$

## Applications

14  $f'(2) = 32$

15  
a 9

b  $y = 9x - 8$

16  
a  $-\frac{1}{2}$

b  $x + 2y - 3 = 0$

17  
a

| Graph     | y-intercept of the tangent line | Gradient of the tangent |
|-----------|---------------------------------|-------------------------|
| $y = x^2$ | $(0, -1)$                       | 2                       |
| $y = x^3$ | $(0, -2)$                       | 3                       |
| $y = x^4$ | $(0, -3)$                       | 4                       |
| $y = x^5$ | $(0, -4)$                       | 5                       |

b  $n$

c No, because then the gradient of the tangent at  $x = 1$  would be 1.

18  
a False

b True

c False

d True

19  
a  $\frac{4}{3}x^{\frac{1}{3}}$

b  $\frac{3}{4}x^{-\frac{1}{4}}$